Vision Documentation

[GitHub]: https://github.com/DanielJ0131/interactive-house/tree/main/SG3_Devices/SG3_Artifacts

Subgroup 3: Devices

Revision History

Date	Version	Description	Author
2026-02-27	1.0	Initial Vision	Tony Nguyen, Patrick Nordahl
2026-04-21	1.1	Refined Problem Statement to address accessibility barriers of rigid physical interfaces; added theoretical goals for personalized biometric authorization and automated emergency health detection (stroke/falls).	Tony Nguyen, Ryad Hazin, Sergej Macut
2026-04-23	1.2	Autonomous Environmental Response Logic	Mustafa Al-Bayati
2026-05-08	1.3	Refined feature set to include Node.js gateway architecture and internal Observer Pattern for decoupled autonomous logic.	Mustafa Al-Bayati, Ryad Hazin,Tony Nguyen, Sergej Macut
2026-05-08	1.4	Updated file to contain Table of content.	Mustafa Al-Bayati, Ryad Hazin,Tony

			Nguyen, Sergej Macut

Product Overview

1. The Problem Statement

- **The problem of:** Testing UbiComp solutions is expensive and potentially unsafe for initial testing with individuals with disabilities. Furthermore, traditional homes lack the "intelligence" to intervene during **acute health crises** (e.g., strokes or falls) when a user cannot physically reach a button or phone.
- **Affects:** Developers in Subgroups 2 and 4, as well as researchers studying human-device communication.
- **The impact of which is:** Slower development cycles and difficulty in verifying how interaction logic (like touch or gesture) affects the actual state of a home.
- **A successful solution would:** Provide a robust **Virtual House logic** that accurately simulates core functions (lights, alarms, temperature) so that web and mobile interfaces can interact with a "real" environment state without needing physical hardware.

2. Product Position Statement

Node.js Gateway Architecture: Transitioned the middleman application to a high-performance **Node.js environment** to handle simultaneous serial and network data with lower latency.

Decoupled Autonomous Logic: Implemented the **Observer Design Pattern** within the C++ firmware, allowing sensors (Subjects) to notify actuators (Observers) directly.

System Responsiveness: This decoupled approach ensures that safety-critical responses—like closing windows during rain—operate independently of external server connectivity or user interface interaction.

3. Stakeholder and User Descriptions

- **Users (Internal):** Subgroup 2 (Controller Interface) will consume your logic to populate their GUIs.
- **Users (End-users):** Individuals with disabilities who will eventually interact with the simulated devices to increase their self-control. Because **traditional physical interfaces are rigid and potentially inaccessible for individuals with limited mobility**, this project provides a safer, simulated environment to test adaptive interaction methods like voice or gesture control.
- **Stakeholders:** HKR University (Academic oversight), UN Sustainable Development Goals (SDG 3, 10, 11 compliance).

4. Key Product Features (Subgroup 3 Focus)

- **Core State Simulation:** Real-time logic for light on/off states and temperature fluctuations.
- **Security Logic:** A functional alarm system simulation that responds to unauthorized triggers.
- **API/Interface Logic:** Providing the "backend" logic so that web and mobile-style GUIs can observe and control the simulated environment.
- **UbiComp Integration:** Supporting the central house server (Subgroup 1) by acting as the primary registry for simulated "Environment" units.
- **Future-State Biometric Integration:** Theoretical support for voice-signature authorization to allow hands-free security control, specifically designed to increase self-control for individuals with disabilities.

- **Personalized Biometric Authorization (Future):** Unlike general voice assistants that respond to any user, this theoretical feature utilizes **unique voice-signatures**. The system verifies the identity of the specific individual to prevent unauthorized disarming by guests or external intruders.

- **Individualized Accessibility:** This allows the house to provide different levels of control based on the specific user's authority level, which is essential for ensuring safety in a shared living environment.

- **Emergency Health Intervention (Future):** A theoretical expansion into bio-state monitoring. By integrating wearable sensors or computer-vision posture analysis via the **Middleman Gateway**, the house can detect anomalies consistent with medical emergencies.

- **Automated Emergency Response:** In the event of a detected stroke, the house logic would automatically execute a "Life-Safety" protocol: unlocking the **Door Servo** for emergency responders, flashing all **LEDs** for visual signaling, and transmitting a high-priority alert to the database.

5. Autonomous Environmental Response Logic

- The system supports intelligent device behavior where the house reacts to environmental changes without human intervention. This is achieved through the **Observer Design Pattern**, which facilitates a one-to-many dependency between sensors and devices, ensuring the firmware maintains safety protocols autonomously even if external communication is lost.

This feature enhances the system by moving from a passive control model to an **active, self-regulating smart environment**.